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NE451 – Lab Assignment 1
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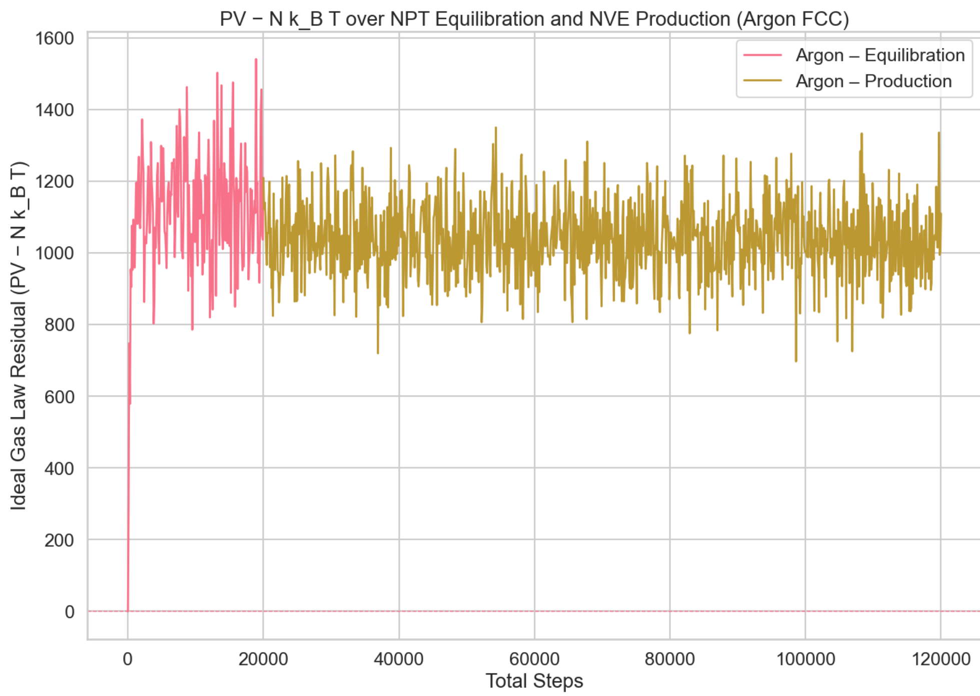
Exercise 1: Thermodynamic Quantity (PV – NkBT)

I created a 4x4x4 FCC lattice with *ase* and chose to use Argon as provided in the example notebook, running with a 1.5 K temp (LJ units), for 20k steps equilibrium and 100k steps sim.



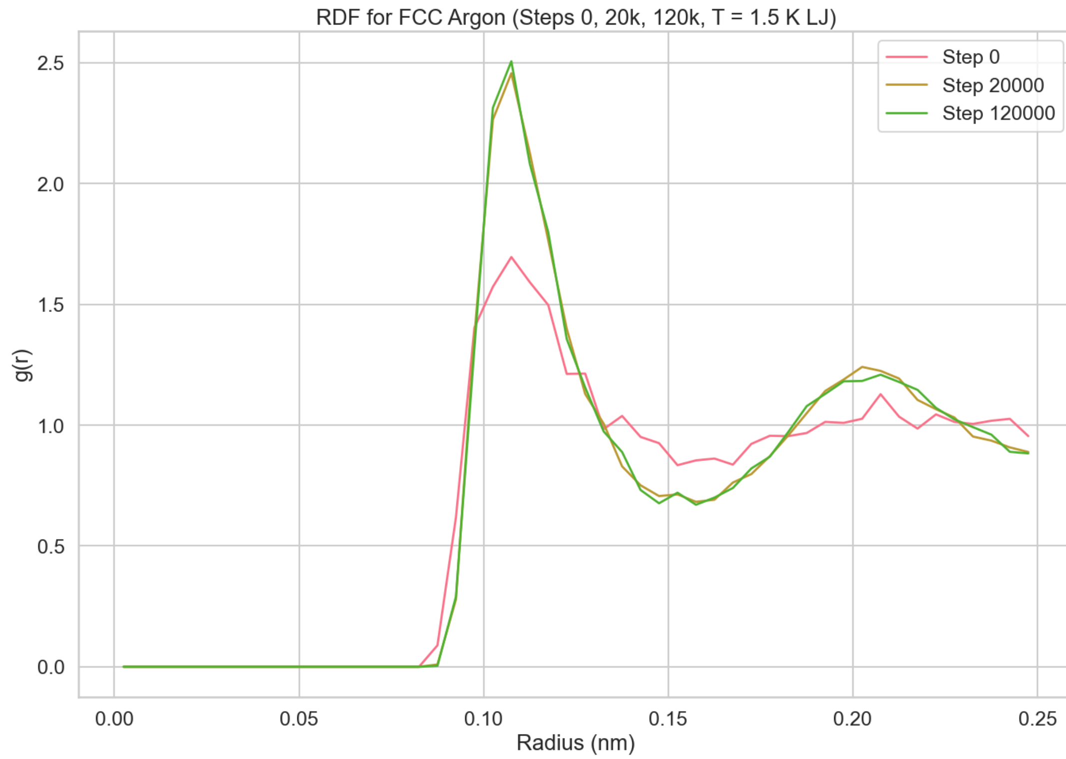
Does PV − NKBT fluctuate? Why?

Yes, but that's expected. In molecular dynamics we're sampling instantaneous microstates of a finite system, not an infinite thermodynamic limit. Because we don't hard-fix both temperature and pressure, their reported values wander in time even when you use thermostats or barostats; those devices only steer the averages toward setpoints. Pressure is especially noisy because it includes contributions from interatomic forces, which spike and relax as atoms collide. On top of that, the velocity-Verlet integrator, while very stable, still introduces tiny numerical errors that add a touch of extra variance. Put together, PV − NKBT naturally jiggles around zero, and the right way to interpret it is via time-averages rather than frame-by-frame values.

How does it compare to the Ideal Gas Law?

In MD, temperature comes from the kinetic energies of the particles simulated, so any small jitter in velocities shows up as temperature noise and propagates into PV − NK_BT. The ideal gas model, by contrast, assumes non-interacting particles and smooth macroscopic averages. Realistic models like Lennard-Jones include interactions, so the instantaneous pressure carries an extra, fluctuating contribution from those forces. At low densities, the long-time average of PV – NK_BT tends to sit close to zero and the system looks "ideal" on average; at higher densities or stronger interactions, you'll see more systematic departures. Either way, the right comparison is on the averaged behavior over a sufficiently long window, not on single snapshots.

Exercise 2: RDF simulation



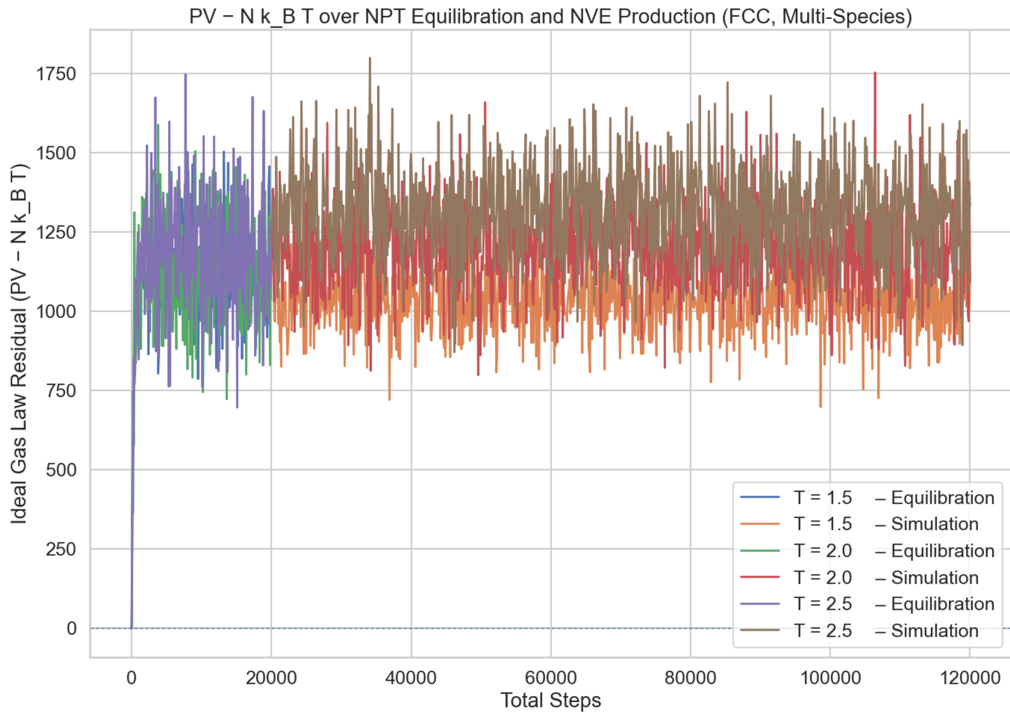
What does the shape of g(r) say about local structure?

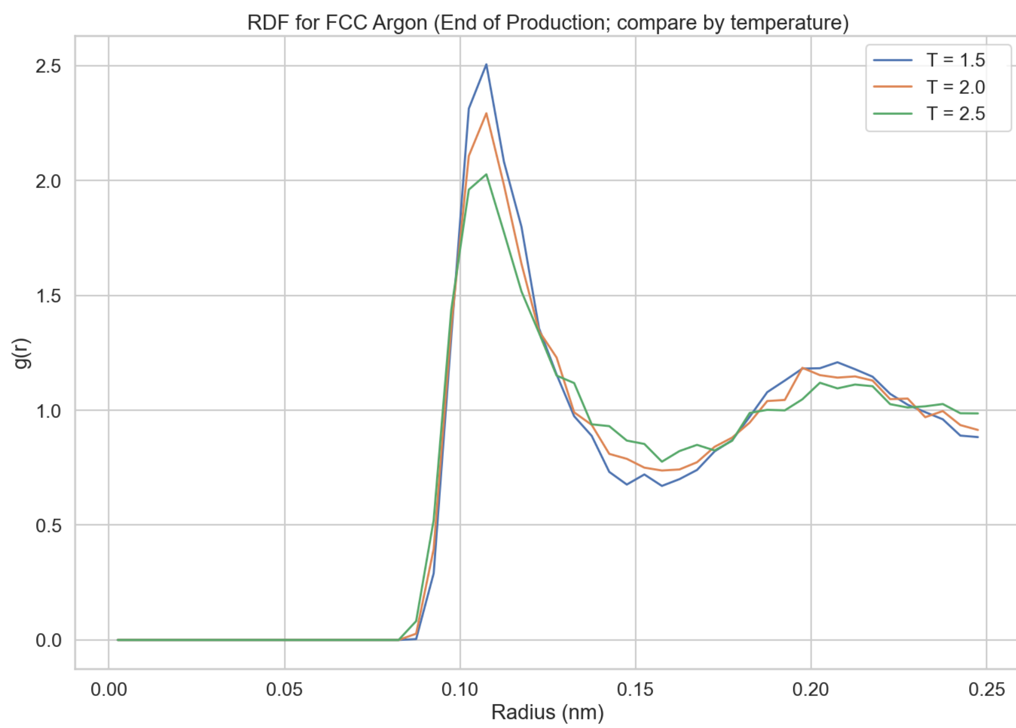
The shape of g(r) reveals how atoms crowd or avoid each other around a reference atom. Values above one mark separations where neighbors are unusually likely; dips mark gaps. The first rise captures the immediate neighborhood; the next rise captures the next layer out. If you integrate g(r) from zero to the first minimum, you obtain the typical count of nearest neighbors i.e., the local coordination number.

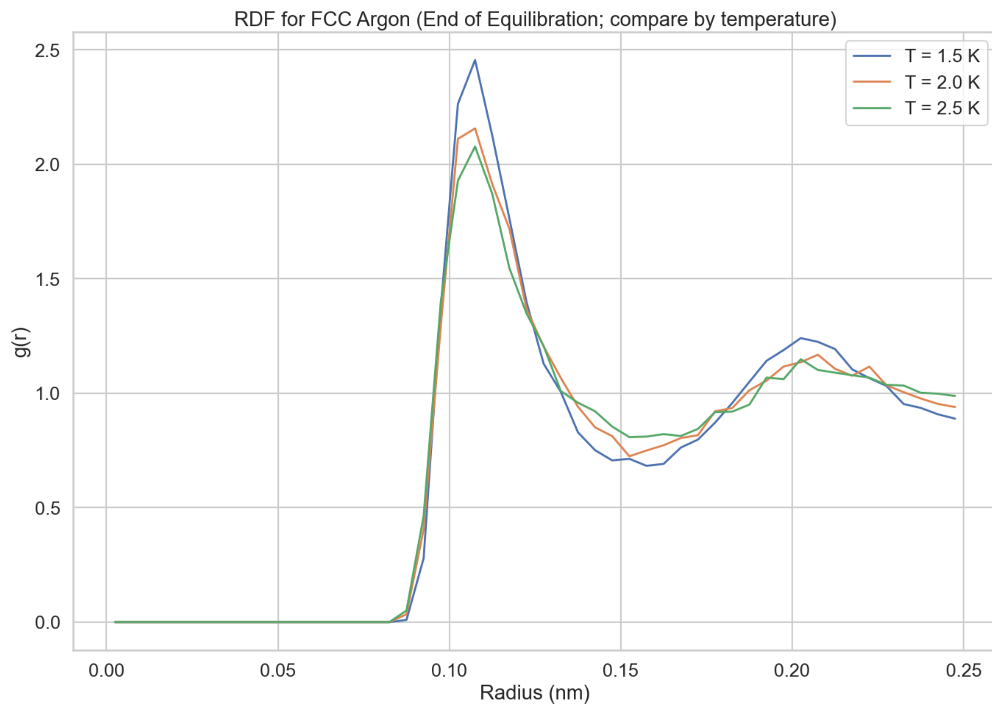
Peaks and their physical meaning:

The first main peak near 0.105 nm is the most likely spacing to the first coordination shell, the closest neighbors. Integrating g(r) up to the first minimum gives the average count of those neighbors. A second peak near 0.205 nm marks the second coordination shell, the next ring of atoms. A taller, sharper second peak means a more defined shell; a lower, broader one means a more liquid-like, less ordered structure.

Exercise 3: Temperature Effects on PV – NkBT and RDF







<u>How does PV − NKBT change with temperature?</u>

Across 1.5 -> 2.5 LJ, the PV − NKBT residual stays near zero and doesn’t change much with T. There’s a slight upward shift in the mean residual as T increases, with a small exception at 2.0 where the average sits a bit lower than the trend. Interpreting that: any increase in T is being offset by how P and V settle (Eq) and evolve (Sim). In other words, the pressure/volume response slightly overcompensates the temperature increase, keeping the residual small and only gently drifting upward.

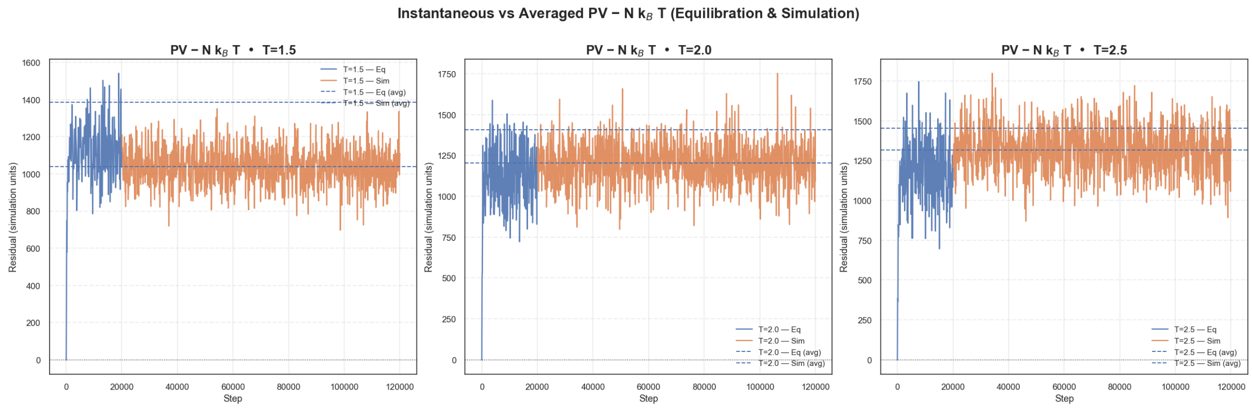
<u>How does the RDF change? Why?</u>

Comparing 1.5, 2.0, 2.5 LJ:

- Peaks get lower and troughs fill in as T rises.
- Short-range order weakens; the distribution flattens toward a more uniform profile.

Why: higher T -> higher kinetic energy -> atoms explore more configurations, so the crystal-like ordering gets blurred. That reduces peak prominence and raises minima, nudging the fluid closer to ideal-gas-like behavior (though not fully ideal at these conditions).

Exercise 4: Average Pressure and Volume



	Run	P_eq	V_eq	T_eq	PV−NkBT_eq(avg)	P_sim	V_sim	T_sim	PV−NkBT_sim(avg)	P_full
0	T=1.5	4.87405	367.98999	1.59752	1384.63775	4.59021	308.95856	1.48167	1038.87783	4.63768
1	T=2.0	4.86286	400.37991	2.11480	1405.60296	5.17297	329.69073	1.96421	1202.64307	5.12111
2	T=2.5	4.91477	431.76579	2.60826	1454.31366	5.53508	358.20002	2.60701	1315.27202	5.43135

The output makes sense. As temperature rises (1.5 -> 2.5 LJ), average pressure and average volume both increases, so the averaged $PV - NK_BT$ residual increases across temperatures. The small deviation at T = 2.0 is reasonable given barostat/thermostat dynamics.

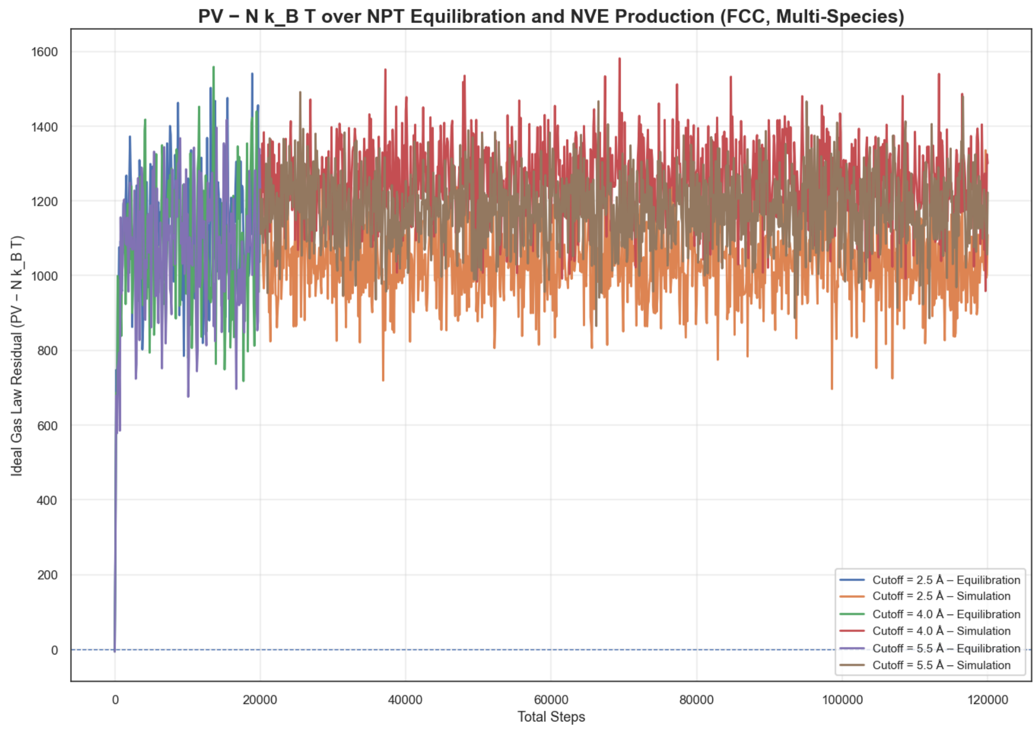
Compare with timestep-by-timestep calculation:

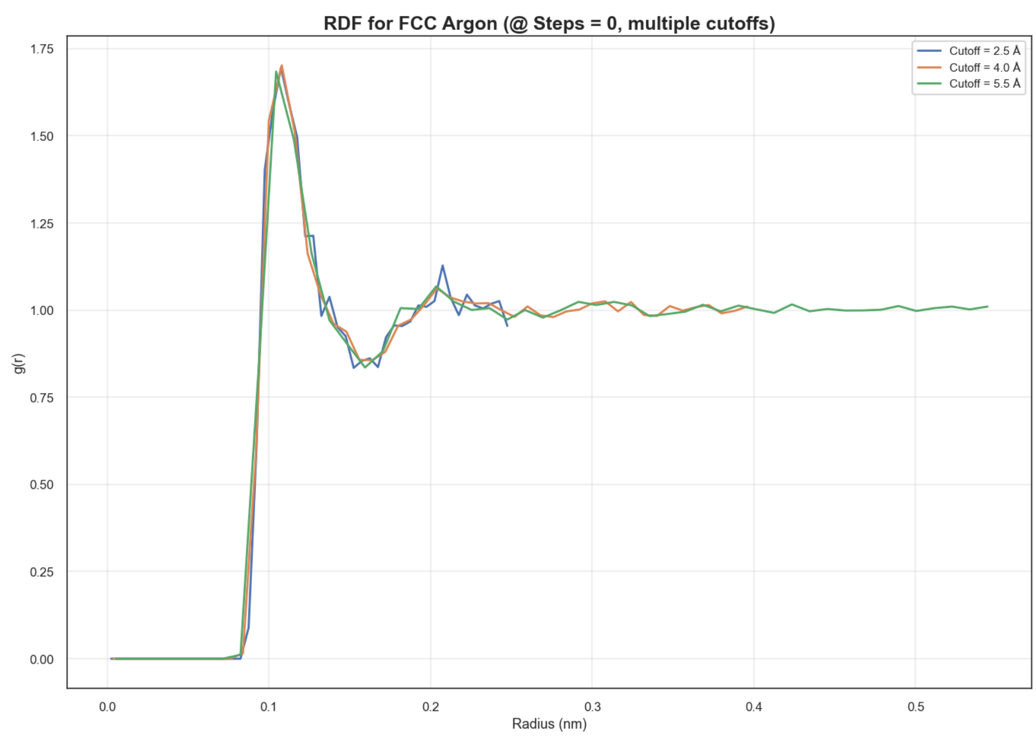
The per-step $PV - NK_BT$ series fluctuates around a fairly steady level, while the "averaged" value is a single number per phase (equilibration or simulation). In the runs, the averaged line sits near the center of the instantaneous curve but not exactly at its time-average, and the baseline shifts upward with temperature. That matches the fact that both mean pressure and mean volume increase with temperature.

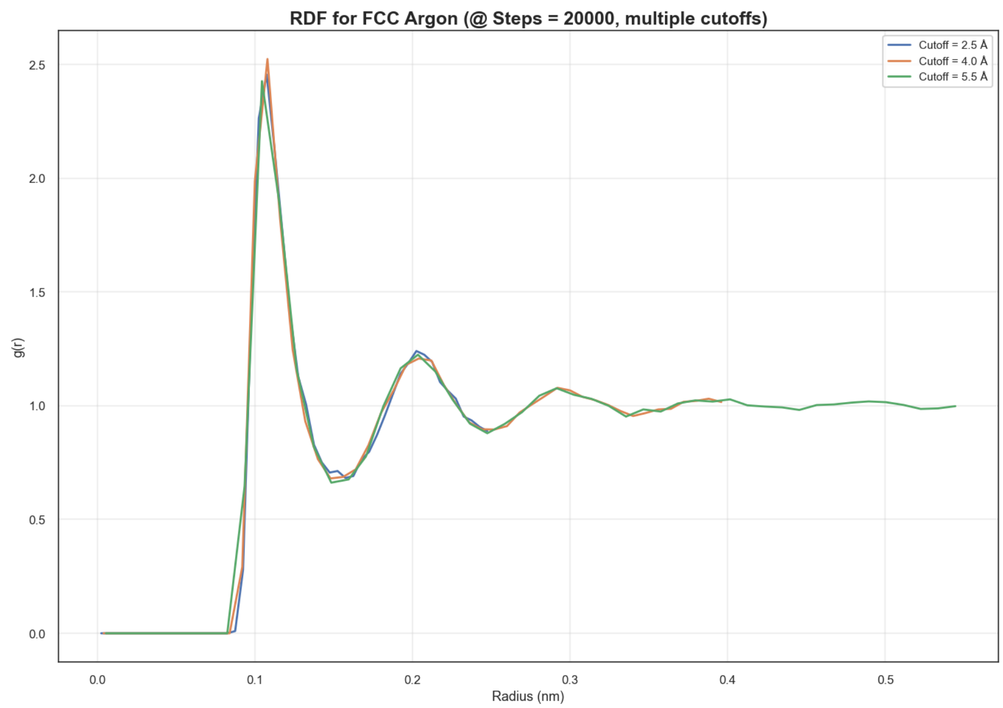
Explain differences between instantaneous and averaged values:

Instantaneous values reflect step-to-step fluctuations and short-term correlations between pressure and volume; when pressure spikes, volume often responds, so their product is not the same as the product of their separate averages. In other words, the average of P*V is generally not equal to (avg P) * (avg V). Slow drift or low-frequency oscillations during equilibration or production also make the per-step series differ from the single averaged number. Overall, the small offsets seen between the instantaneous series and the averaged calculation are expected.

Exercise 5: Effect of Interaction Potential







<u>How do results change with the cutoff?</u>

Across the three cutoffs (2.5 Å, 4.0 Å, 5.5 Å) the PV − NKBT traces are basically the same. You can see tiny shifts in the average level and a bit of noise change, but no qualitative change in behavior. That's expected for LJ at these state points: most of the contribution to forces and virial comes from nearer neighbors, and the 1/r^6 tail beyond a few σ adds only a small correction. The clear difference is wall-clock time: as you increase the cutoff, the neighbor list grows roughly with rc^3, so the compute cost climbs even though the physics barely moves.

<u>Any differences in RDF? Why?</u>

Not really much. Peak positions and overall shape are essentially unchanged; at larger cutoff you may notice the curve extends to larger r and looks slightly smoother. Reason: adding distant neighbors doesn't reorganize the local structure; RDF is dominated by the first few shells so g(r) inside those shells hardly changes. The smoother look at larger rc comes from having more pairs (better statistics) and, depending on binning, a mild reduction in truncation artifacts.

Exercise 6: Mean Squared Displacement and Diffusion Coefficient

Is MSD linear with time? What does it indicate?

After the initial equilibration, the mean squared displacement (MSD) increases proportionally with time. This linear region reflects the transition into the diffusive regime: atoms undergo random thermal motion, and the system exhibits normal diffusion. The slope of this MSD–time curve corresponds to the diffusion coefficient, and once the line settles into a steady gradient it shows that the diffusion constant has become well-defined for the system.

Compare diffusion for different atom types:

When comparing species, the general trend is that heavier atoms diffuse more slowly because their larger mass reduces mobility. In practice, the MSD curves reveal distinct slopes: lighter Neon exhibits a steeper slope (larger D), Argon is intermediate, and Krypton shows the slowest growth. This outcome matches physical expectations, since bigger atoms experience stronger Lennard-Jones attractions and require more thermal energy to move through the medium. The differences in MSD growth rates therefore highlight the mass and interaction-dependent nature of diffusion.